

Course Syllabus SMS 491

Field Studies in Ocean and Climate Science

Fall 2026 - 3 Credits

Meeting Time: Mondays and Wednesdays 1:00 - 1:50 PM EST

Meeting Location: In-person 354 Aubert Hall

Class Size: 10 students

Instructor: Tiago Carrilho Biló

Contact: tiago.carrilho@maine.edu

TA: Andrew Treat (andrew.treat@maine.edu)

Office Hours: By appointment, in-person 454 Aubert Hall

Course Description

This course provides an intensive, hands-on research experience for students from diverse majors, including social work, psychology, engineering, earth science, and biology, and includes a 5-day trip to the Hurricane Island Center for Science and Leadership in ME before the Fall semester starts. During the course, the students will be introduced to interdisciplinary environmental research and monitoring techniques focused on the Gulf of Maine; receive hands-on training in ocean and climate field methods; develop skills in data collection, analysis, interpretation, and science communication; and foster teamwork, leadership, and problem-solving in real-world research settings.

Learning Objectives

By the end of the course, students will be able to:

- Design and propose a small-scale research project integrating marine mammal acoustic and/or oceanographic data.
- Demonstrate effective teamwork and leadership skills in both field and classroom settings.
- Apply fundamental field methods and analytical tools to collect, process, and interpret environmental data.
- Communicate scientific findings to both scientific and non-specialist audiences through a final report.

Prerequisites

There are no prerequisites for this course. Students must apply for the course and be accepted by the course instructor to enroll.

Classroom Conduct

I am committed to cultivating an inclusive, respectful, and supportive learning environment for everyone regardless of background, identity, appearance, or communication style. Disrespectful language or behavior, including anything that marginalizes, threatens, or makes others feel unwelcome, will not be tolerated. To help maintain a positive and professional atmosphere, all students, instructors, and guests are encouraged to:

- Communicate in a kind and inclusive manner
- Respect diverse perspectives and lived experiences
- Receive and offer feedback with openness and humility
- Engage with peers and instructors in a thoughtful and courteous way

Technology and Personal Gear

This course will involve data analysis, research, and scientific writing. Therefore, students are expected to have a laptop with Microsoft suite (Word, Powerpoint, Excel) installed or access to the university systems.

For the field component of the course, students are expected to bring personal gear to be comfortable while in the field. This includes warm and waterproof clothing (e.g., rain jacket, down/fleece jacket, hat, gloves). Selected students are expected to be able to meet via zoom for an informative session about the field trip to ensure everyone is ready and equipped with what they need.

Course Components

- **Weekly Meetings (1 hour and 40 mins):** Seminar-style discussions, project planning, and skill-building workshops for data analysis and scientific writing.
- **Field Component:** 5-day-long trip to Hurricane Island with instrument recovery and deployment aboard a research vessel in the Gulf of Maine.
- **Project Work:** Students will collaboratively design a project, analyze data, and develop a final report.
- **Final Deliverables:** A written report and an outreach product (e.g., blog post, infographic, art piece, or community presentation).

Course Schedule and Assignments

- Field Trip: August 25-29
- Report Outline: Project theme and report “skeleton”
- Outreach Product or Presentation
- Peer Review and Feedback report
- Final Report

Evaluation and Grading

- Attendance & Participation (1 class/week) – 20%
- Field Preparation & Participation – 30%
- Report Outline – 10%
- Peer Review & Feedback – 10%
- Outreach Product or Presentation – 10%
- Final Report – 20%

GRADING SUMMARY							
A	93 - 100	B	83 - 86	C	73 - 76	D	63 - 66
A –	90 - 92	B –	80 - 82	C –	70 - 72	D –	60 - 62
B +	87 - 89	C +	77 - 79	D +	67 - 69	F	< 60

Course Policies

- **Attendance:** Absences must be communicated in advance.
- **Collaboration:** Central to course success; students are expected to work equitably and respectfully in teams.
- **Academic Integrity:** Applies to all written, visual, and presented work.
- **Accommodations:** Available through [University Accessibility Services].
- **Late Policy:** Assignments will be accepted up to a week late, but will be deducted -10% for every day it is late.

Generative AI tools

You may use generative AI tools on assignments in this course as you see fit. If you choose to use generative AI tools, please remember that they are typically trained on limited datasets that may be out of date. Additionally, generative AI datasets are trained on pre-existing material, including copyrighted material; therefore, relying on a generative AI tool may result in plagiarism, copyright violations, or wrong information. It is your responsibility—not the tool's—to assure the quality, integrity, and accuracy of work you submit in any college course. If you do use generative AI tools on the homework assignments in this class, you must properly document and credit the tools themselves. Cite the tool you used, following the pattern for computer software given in the specified style guide. Additionally, please include a brief (one sentence) description of how you used the tool followed by a list of the prompts used (bullet points).

Campus Policies

Academic Honesty Statement

Academic honesty is very important. It is dishonest to cheat on exams, to copy term papers, to submit papers written by another person, or generated by software or systems without the explicit approval of the instructor, to fake experimental results, or to copy or reword parts of books or articles into your own papers without appropriately citing the source. Students committing or aiding in any of these violations may be given failing grades for an assignment or for an entire course, at the discretion of the instructor. In addition to any academic action taken by an instructor, these violations are also subject to action under the University of Maine Student Conduct Code. The maximum possible sanction under the student conduct code is dismissal from the University. Please see the University of Maine System's Academic Integrity Policy listed in the Board Policy Manual as Policy 314 (***Date Issued:** September 1, 2020): <https://www.maine.edu/board-of-trustees/policy-manual/section-314/>

Last Updated: 07.07.2023

Course Schedule Disclaimer (Disruption Clause)

In the event of an extended disruption of normal classroom activities (due to COVID-19 or other long-term disruptions), the format for this course may be modified to enable its completion within its programmed time frame. In that event, you will be provided an addendum to the syllabus that will supersede this version.

Last Updated: 08.17.2020

Observance of Religious Holidays/Events

The University of Maine recognizes that when students are observing significant religious holidays, some may be unable to attend classes or labs, study, take tests, or work on other assignments. If they provide adequate notice (at least one week and longer if at all possible), these students are allowed to make up course requirements as long as this effort does not create an unreasonable burden upon the instructor, department or University. At the discretion of the instructor, such coursework could be due before or after the examination or assignment. No adverse or prejudicial effects shall result to a student's grade for the examination, study, or course requirement on the day of religious observance. The student shall not be marked absent from the class due to observing a significant religious holiday. In the case of an internship or clinical, students should refer to the applicable policy in place by the employer or site.

Last Updated: 08.17.2020

Sexual Discrimination Reporting

The University of Maine is committed to making campus a safe place for students. Because of this commitment, if you tell any of your teachers about sexual discrimination involving members of the campus, **your teacher is required to report** this information to Title IX Student Services or the Office of Equal Opportunity.

Behaviors that can be “sexual discrimination” include sexual assault, sexual harassment, stalking, relationship abuse (dating violence and domestic violence), sexual misconduct, and gender discrimination. Therefore, all of these behaviors must be reported.

Why do teachers have to report sexual discrimination?

The university can better support students in trouble if we know about what is happening. Reporting also helps us to identify patterns that might arise – for example, if more than one victim reports having been assaulted or harassed by the same individual.

What will happen to a student if a teacher reports?

An employee from Title IX Student Services or the Office of Equal Opportunity will reach out to you and offer support, resources, and information. You will be invited to meet with the employee to discuss the situation and the various options available to you.

If you have requested confidentiality, the University will weigh your request that no action be taken against the institution’s obligation to provide a safe, nondiscriminatory environment for all students. If the University determines that it can maintain confidentiality, you must understand that the institution’s ability to meaningfully investigate the incident and pursue disciplinary action, if warranted, may be limited. There are times when the University may not be able to honor a request for confidentiality because doing so would pose a risk to its ability to provide a safe, nondiscriminatory environment for everyone. If the University determines that it cannot maintain confidentiality, the University will advise you, prior to starting an investigation and, to the extent possible, will share information only with those responsible for handling the institution’s response

The University is committed to the well-being of all students and will take steps to protect all involved from retaliation or harm.

If you want to talk in confidence to someone about an experience of sexual discrimination, please contact these resources:

*For confidential resources on campus: **Counseling Center: 207-581-1392** or **Northern Light Primary Care, University of Maine: at 207-581-4000**. Confidential Resource Advisor: 207-571-5372 (call or text). Or see the [Confidential Resource Advisor website](#) for a complete list of services and resources (open in a new window).*

For *confidential resources off campus*: **Rape Response Services**: 1-800-871-7741 or **Partners for Peace**: 1-800-863-9909.

Other resources: The resources listed below can offer support but may have to report the incident to others who can help:

For *support services on campus*: **Title IX Student Services**: 207-581-1406, **Office of Community Standards**: 207-581-1409, **University of Maine Police**: 207-581-4040 or 911. Or [see the Title IX Student Services website for a complete list of services \(opens in a new window\)](#). Also, [Student Wellness Resource Center \(opens in a new window\)](#).

Last Updated: 07.31.2024

Students Accessibility Services Statement:

If you have a disability for which you may be requesting an accommodation, please contact Student Accessibility Services, located at the Center for Accessibility and Volunteer Engagement at the UCU, 139 Rangeley Rd, um.sas@maine.edu, 207.581.2319, as early as possible in the term. Students may begin the accommodation process by submitting an accommodation request form online and uploading documentation at https://umaine-accommodate.symplicity.com/public_accommodation/. Once students meet with SAS and eligibility has been determined, students submit an online request with SAS each semester to activate their approved accommodations. SAS creates an accessibility letter each semester which informs faculty of potential course access and approved reasonable accommodations; the letter is sent directly to the course instructor. Students who have already been approved for accommodations by SAS and have a current accommodation letter should meet with me (Dr. Tiago Carrilho Bilo) privately as soon as possible.

Last Updated: 04.01.2024